Block practice · Trigonometric identities and equations

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Reading a sine model: values, biggest and smallest

Exact values with the addition formulae

Getting one exact trig ratio from another

OLevels.com SCHOOL

Practice workbook · Reading a sine model: values, biggest and smallest

Reading off values of a sinusoidal model or graph by substituting and using exact values and the range of sine and cosine.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-01

1. Question 1 · part b.ii · 2 marks

2020 NOV · Paper 01 (regional R)

Worked-example retry · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

1 Here is a formula

$$P = 3 + 2\sin\left(\frac{3\pi t}{8}\right) \qquad 0 \leqslant t \leqslant 12$$

(a) Find the exact value of P when $t = \frac{10}{3}$

(2)

(b) Find

(i) the largest value of P

(ii) the smallest value of P

(2)

(c) Find the least value of t for which $P = 4$

(3)

OLevels.com SCHOOL

1. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 1 · part a · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

1 Here is a formula

$$P = 3 + 2\sin\left(\frac{3\pi t}{8}\right) \qquad 0 \leqslant t \leqslant 12$$

(a) Find the exact value of P when $t = \frac{10}{3}$

(2)

(b) Find

(i) the largest value of P

(ii) the smallest value of P

(2)

(c) Find the least value of t for which $P = 4$

(3)

OLevels.com SCHOOL

2. Question 1 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 1 · part b.i · 0 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

1 Here is a formula

$$P = 3 + 2\sin\left(\frac{3\pi t}{8}\right) \qquad 0 \leqslant t \leqslant 12$$

(a) Find the exact value of P when $t = \frac{10}{3}$

(2)

(b) Find

(i) the largest value of P

(ii) the smallest value of P

(2)

(c) Find the least value of t for which $P = 4$

(3)

OLevels.com SCHOOL

3. Question 1 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 12 · part a · 1 mark

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

12

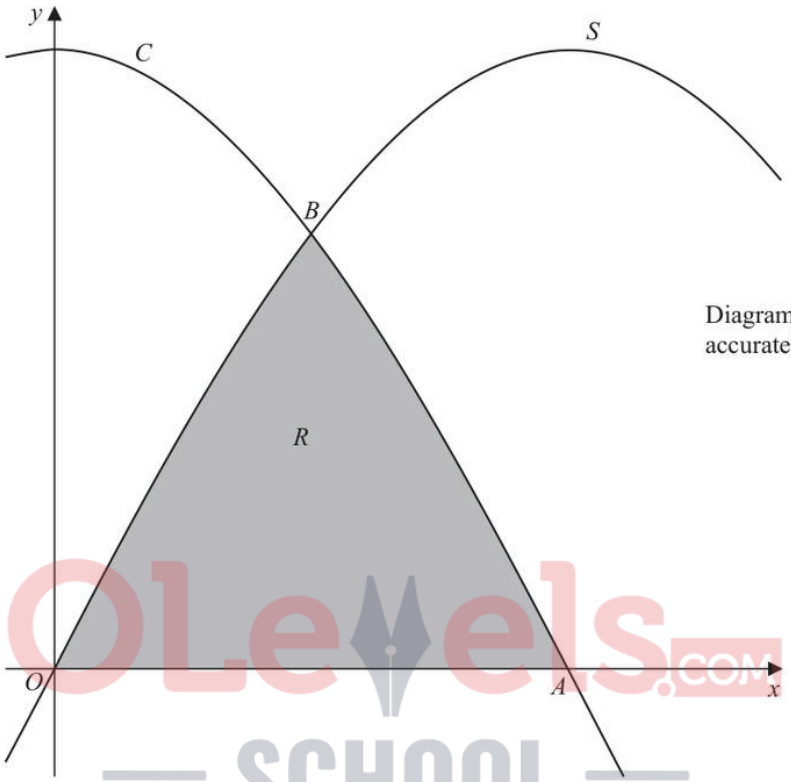


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the x-axis, the curve S with equation $y = 2\sin x$ and the curve C with equation $y = 2\cos x$.
As shown in Figure 2, C crosses the x-axis at the point A.

(a) Write down the x coordinate of A.

(1)

As shown in Figure 2, C and S intersect at the point B.

(b) Find the x coordinate of B.

(2)

(c) Using calculus, find the area of the shaded region R.
Give your answer in the form $a - \sqrt{b}$ where a and b are integers.

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

38

P 6 2 2 8 3 A 0 3 8 4 0

4. Question 12 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Exact values with the addition formulae

Finding exact trig values of angles such as 15, 75, 105 or 255 degrees by writing them as sums or differences of 30, 45 and 60 degrees.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-21

1. Question 9 · part a · 3 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part a only.

9

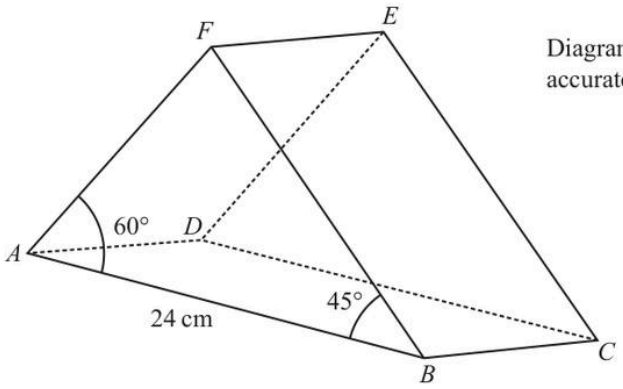


Figure 5

Figure 5 shows a right triangular prism $ABCDEF$ where $ABCD$ is a rectangle.

$AF = DE$ $BF = CE$ $AD = FE = BC$ $AB = DC = 24$ cm

$\angle ABF = \angle DCE = 45°$ $\angle BAF = \angle CDE = 60°$

Using a formula from page 2,

(a) show that $\sin AFB = \dfrac{\sqrt{2} + \sqrt{6}}{4}$

(3)

Without using a calculator,

(b) show that $BF = 12\left(3\sqrt{2} - \sqrt{6}\right)$ cm

(5)

The angle between the plane AEB and the plane $ABCD$ is 65°

(c) Find, in cm to 2 significant figures, the length of EF

(3)

(d) Find, in degrees to one decimal place, the size of the angle between the line CF and the plane $ABCD$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 6 5 0 8 A 0 2 2 3 2

1. Question 9 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part b · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

2 Angle α is acute such that $\cos\alpha = \frac{3}{5}$

Angle β is obtuse such that $\sin\beta = \frac{1}{2}$

(a) Find the exact value of

(i) $\tan\alpha$

(ii) $\tan\beta$

(3)

(b) Hence show that

$$\tan(\alpha+\beta) = \frac{m\sqrt{3}-n}{n\sqrt{3}+m}$$

where m and n are positive integers whose values are to be found.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 2 · part b · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 2 is 6 marks)

2. Question 2 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 11 · part c.i · 2 marks

2020 JAN · Paper 02

Further practice · Attempt part c.i only.

11 (a) Express the equation

$$3\sin(A - B) = \sin(A + B)$$

in the form $\tan A = k \tan B$, giving the value of the integer k.

(4)

(b) Given that $\theta \neq \frac{(2n + 1)\pi}{2}$ where $n \in \mathbb{Z}$,

show that $\frac{\cos^4 \theta - \sin^4 \theta}{\cos^2 \theta} = 1 - \tan^2 \theta$

(3)

(c) Using the exact values of $\sin x°$, $\cos x°$ and $\tan x°$ for $x = 30, 45, 60$

show that

(i) $\cos 15° = \frac{\sqrt{6} + \sqrt{2}}{4}$

(2)

(ii) $\tan 255° = \frac{3 + \sqrt{3}}{3 - \sqrt{3}}$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

3. Question 11 · part c.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 11 · part c.ii · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c.ii only.

11 (a) Express the equation

$$3\sin(A - B) = \sin(A + B)$$

in the form $\tan A = k \tan B$, giving the value of the integer k.

(4)

(b) Given that $\theta \neq \frac{(2n + 1)\pi}{2}$ where $n \in \mathbb{Z}$,

show that $\frac{\cos^4 \theta - \sin^4 \theta}{\cos^2 \theta} = 1 - \tan^2 \theta$

(3)

(c) Using the exact values of $\sin x°$, $\cos x°$ and $\tan x°$ for $x = 30, 45, 60$

show that

(i) $\cos 15° = \frac{\sqrt{6} + \sqrt{2}}{4}$

(2)

(ii) $\tan 255° = \frac{3 + \sqrt{3}}{3 - \sqrt{3}}$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

4. Question 11 · part c.ii · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c.ii only.

Question 11 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 11 · part c.ii · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c.ii only.

Question 11 continued

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 11 is 13 marks)

4. Question 11 · part c.ii · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Getting one exact trig ratio from another

Using sin^2 + cos^2 = 1 (and tan = sin/cos) to turn one known trig ratio of an angle into the exact value of another, with the right sign.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-02

1. Question 2 · part a · 4 marks

2019 JUN · Paper 01 (regional R)

Worked-example retry · Attempt part a only.

2

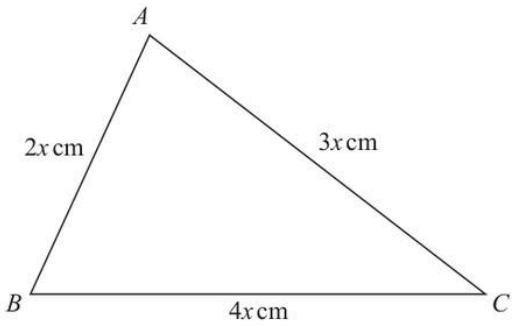


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle ABC in which

$AB = 2x$ cm $\quad AC = 3x$ cm $\quad BC = 4x$ cm

(a) Show that $\sin ABC = \frac{3\sqrt{15}}{16}$

(4)

Given that the area of triangle ABC is $\frac{75\sqrt{15}}{64}$ cm^2

(b) find the value of x.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 2 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part a.i · 0 marks

2021 JUN · Paper 01

Further practice · Attempt part a.i only.

2 Angle α is acute such that $\cos\alpha = \frac{3}{5}$

Angle β is obtuse such that $\sin\beta = \frac{1}{2}$

(a) Find the exact value of

(i) $\tan\alpha$

(ii) $\tan\beta$

(3)

(b) Hence show that

$$\tan(\alpha + \beta) = \frac{m\sqrt{3} - n}{n\sqrt{3} + m}$$

where m and n are positive integers whose values are to be found.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 2 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 2 · part a.ii · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part a.ii only.

2 Angle α is acute such that $\cos\alpha = \frac{3}{5}$

Angle β is obtuse such that $\sin\beta = \frac{1}{2}$

(a) Find the exact value of

(i) $\tan\alpha$

(ii) $\tan\beta$

(3)

(b) Hence show that

$$\tan(\alpha + \beta) = \frac{m\sqrt{3} - n}{n\sqrt{3} + m}$$

where m and n are positive integers whose values are to be found.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 6 · part a.ii · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part a.ii only.

6 In triangle ABC, $AC = x$ cm, $AB = (x + 3)$ cm and $\angle ABC = 30°$

Given that $\angle ACB = \theta°$ where $0 < \theta < 90$

(a) show that

(i) $\sin\theta° = \dfrac{x + 3}{2x}$

(ii) $\cos\theta° = \dfrac{\sqrt{3x^2 - 6x - 9}}{2x}$

(5)

Given that the size of $\angle BAC$: the size of $\angle ABC = 7 : 2$

(b) find the exact value of x

Give your answer in the form $a + a\sqrt{b}$ where a and b are prime numbers.

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 6 · part a.ii · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL